



**FACULTY OF COMMUNICATION, VISUAL ARTS AND COMPUTING
COMPUTING DEPARTMENT**

ICS1253 IOT AND CLOUD COMPUTING

TITLE: SMART GAS LEAKAGE MONITORING SYSTEM USING
IOT AND CLOUD PLATFORM

PROGRAMME: BACHELOR IN INFORMATION TECHNOLOGY (HONS.)

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1. Introduction

The rapid advancement of Internet of Things (IoT) technology has significantly transformed the way environmental monitoring systems are designed and implemented. IoT enables physical devices embedded with sensors, software, and communication modules to collect and exchange data through internet connectivity. This integration enhances automation, real-time monitoring, and remote accessibility across various industries.

Gas leakage remains one of the most critical safety hazards in residential buildings, laboratories, commercial kitchens, and industrial environments. Harmful gases such as carbon dioxide and volatile organic compounds may accumulate without immediate human detection, potentially leading to fire accidents, explosions, and long-term health complications. Traditional gas detection systems are typically limited to standalone alarm devices that provide only local alerts and do not support remote monitoring or historical data analysis.

To overcome these limitations, this project develops a Smart Gas Leakage Monitoring System using IoT and cloud computing technology. The system integrates an MQ135 gas sensor with an ESP32 microcontroller featuring built-in WiFi capability. The collected sensor data is transmitted to the ThingSpeak cloud platform for real-time visualization and storage. By combining hardware integration, embedded programming, wireless communication, and cloud analytics, this project demonstrates a practical implementation of IoT in environmental safety monitoring.

2. Problem Statement

Gas leakage is a serious safety issue in residential, laboratory, commercial, and industrial environments. Harmful gases can accumulate without immediate detection, potentially causing fire accidents, explosions, and long-term health risks. The absence of early detection mechanisms increases the likelihood of severe damage and delayed emergency response.

Currently, many locations still rely on manual inspection or standalone gas detectors that only provide local audible alerts. These traditional systems do not support remote monitoring, real-time data transmission, or cloud-based data storage. As a result, users cannot monitor environmental conditions when they are away, and historical data cannot be analyzed to identify trends or recurring leakage patterns.

Therefore, there is a need for an IoT-based smart gas monitoring system that can continuously detect gas concentration levels, provide immediate alerts, and transmit data to a cloud platform for real-time remote monitoring and analysis.

3. Project Objectives

The main objectives of this project are:

- I. To design and develop a Smart Leakage Monitoring System using IoT technology for real-time gas detection.
- II. To monitor gas concentration levels continuously using the MQ135 sensor and activate an alert mechanism when the readings exceed predefined safety thresholds.
- III. To transmit gas concentration data to a cloud platform for real-time remote monitoring and basic data analysis.

4. Hardware Setup

The hardware architecture of the proposed IoT system is centered around the ESP32 development board, which functions as the main processing unit. The ESP32 was selected due to its integrated WiFi capability, sufficient processing power, low energy consumption, and compatibility with Arduino IDE programming. Unlike traditional designs that require separate WiFi modules, the ESP32 simplifies circuit complexity by providing built-in wireless communication support.

The MQ135 gas sensor is connected to the analog input pin of the ESP32. This sensor is capable of detecting various gases such as ammonia, smoke, benzene, and carbon dioxide. It outputs analog voltage values that correspond to gas concentration levels in the surrounding air.

To enhance real-time user awareness, two LED indicators were incorporated into the system design. The green LED indicates safe environmental conditions, while the red LED signals hazardous gas concentration levels. Each LED is connected through a 220-ohm resistor to regulate current flow and prevent component damage.

An active 5V buzzer is also integrated to provide immediate audible alerts when gas levels exceed a predefined threshold value. The combination of sensor, microcontroller, visual indicators, and audible alarm creates a complete embedded safety monitoring unit.

Fritzing Sketch Diagram:

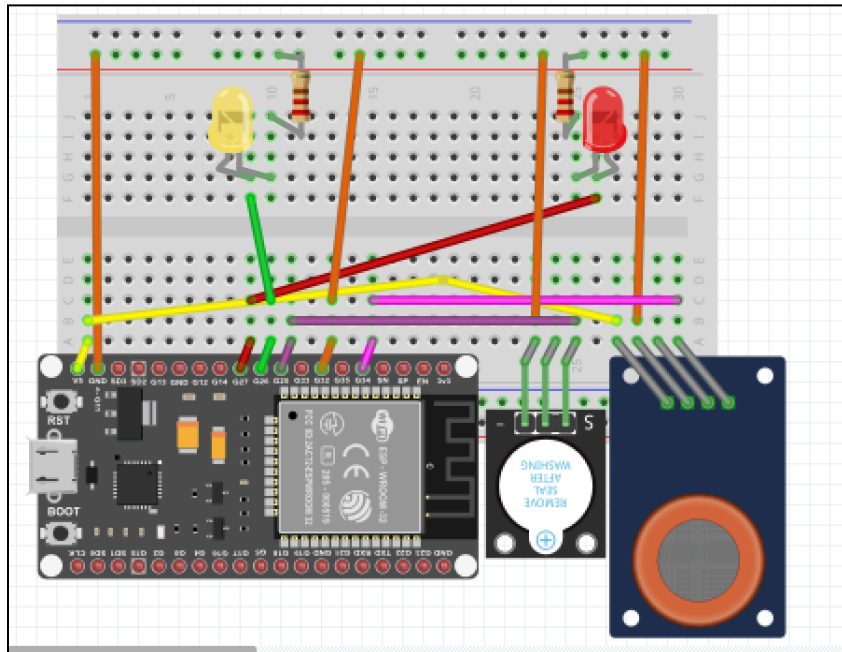


Figure 4.1 Fritzing Sketch

ESP-32 :

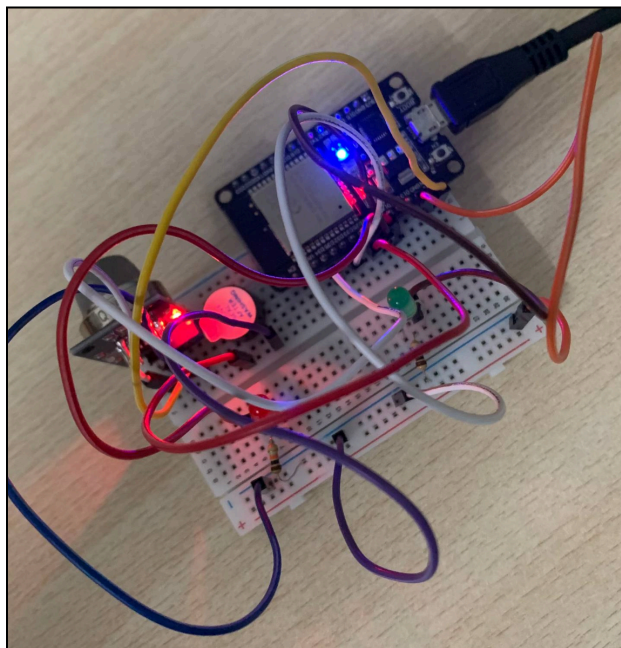


Figure 4.2 ESP-32

5. Software and Communication

The system firmware was developed using Arduino C/C++ programming language within the Arduino IDE environment. The software architecture follows a continuous loop structure where sensor readings are collected, processed, and transmitted periodically.

The ESP32 reads analog data from the MQ135 sensor and converts the raw signal into a measurable gas concentration value. The firmware then compares the sensor reading with a predefined safety threshold value of 600. If the gas concentration remains below the threshold, the green LED remains activated while the red LED and buzzer stay turned off.

However, when the gas concentration exceeds the threshold, the system immediately activates the red LED and buzzer to indicate a dangerous condition. This threshold-based decision logic ensures fast and automatic response without human intervention.

Simultaneously, the ESP32 establishes a WiFi connection and transmits the gas concentration data to the ThingSpeak cloud server using HTTP protocol. Data transmission occurs at 15-second intervals to ensure real-time updates while complying with the cloud platform limitations.

6. Cloud Integration

ThingSpeak was selected as the cloud platform for this project due to its simplicity, support for HTTP data transmission, and built-in data visualization tools. The ESP32 sends sensor readings to a designated ThingSpeak channel using an API key for authentication.

Each transmitted reading is stored along with an automatic timestamp, enabling the formation of time-series data. The platform provides graphical line charts that display gas concentration trends in real time.

This cloud integration enables remote monitoring, allowing users to observe environmental conditions from any location with internet access. Additionally, historical data storage allows pattern identification and anomaly detection, which are not possible in traditional standalone detection systems.

The integration of cloud computing transforms the project into a complete IoT ecosystem rather than a simple embedded alarm system.

7. Sensor Data Monitoring and Testing

To evaluate system performance, controlled experiments were conducted under two primary conditions: normal air environment and gas exposure environment.

Before testing, the MQ135 sensor was allowed to warm up for several minutes to stabilize baseline readings. Under normal environmental conditions, sensor readings fluctuated between 555 and 598, remaining consistently below the safety threshold of 600. During this condition, only the green LED remained active, and no buzzer activation occurred.

For gas exposure testing, alcohol-based perfume was introduced near the sensor. The gas concentration values increased significantly, exceeding the threshold and reaching peak values around 685. Once the threshold was exceeded, the red LED and buzzer were activated immediately. The ThingSpeak dashboard reflected a noticeable spike in the graphical data representation, confirming successful real-time transmission and cloud synchronization.

8. Data Analysis

In this project, two tests were conducted: Normal Condition Test and Gas Exposure Test. During the Normal Condition Test, gas readings were below 600 and categorized as safe. During the Gas Exposure Test, readings exceeded 600, indicating unsafe air conditions. These results show that the system can differentiate between safe and hazardous environments.

Normal Condition Test:

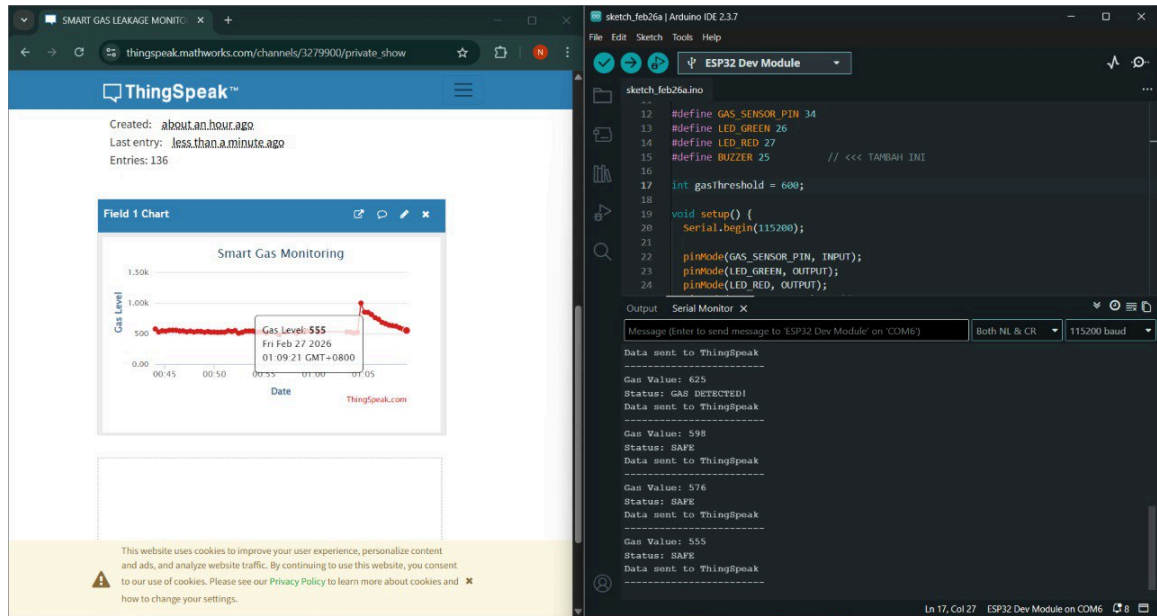


Figure 8.1 Normal Condition Test

Figure 8.1 shows the system performance under normal environmental conditions without any gas exposure. Based on the ThingSpeak graph, the gas readings fluctuated between approximately 550 and 600. These values represent the baseline air quality in the surrounding environment. The readings remained stable without any sudden spikes, indicating that no abnormal gas concentration was detected during this period.

The Serial Monitor output further supports this observation, where gas values such as 555, 576, and 598 were recorded. Since all readings remained below the predefined threshold value of 600, the system categorized the condition as “SAFE.” Under this condition, only the green LED was activated, while the red LED and buzzer remained turned off.

The consistent readings shown in both the Serial Monitor and ThingSpeak dashboard confirm that the system operates accurately under safe conditions. The absence of false alarms demonstrates that the threshold configuration is appropriate and that the sensor is stable when detecting normal air quality levels.

Gas Exposure Test:

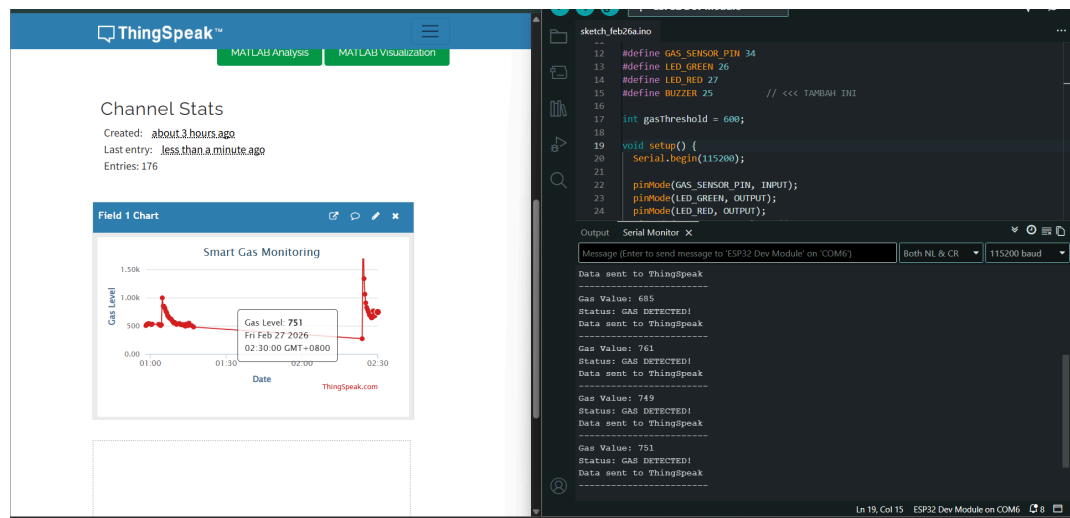


Figure 8.2 Gas Exposure Test

Figure 8.2 shows the system performance during the gas exposure test. In this test, an external gas source was introduced near the MQ135 sensor to simulate unsafe air conditions. Based on the ThingSpeak graph, a significant spike in gas concentration can be observed, where the readings increased sharply above the normal baseline level. The values exceeded the predefined threshold of 600, reaching readings around 685, 749, and 761.

The Serial Monitor output confirms this increase, displaying the status “GAS DETECTED!” once the readings surpassed the threshold value. This indicates that the system successfully identified abnormal gas concentration levels. Upon exceeding the threshold, the red LED and buzzer were activated automatically as part of the alert mechanism, providing both visual and audible warnings.

The sharp increase in the graph followed by gradual stabilization demonstrates the sensitivity and responsiveness of the MQ135 sensor. The synchronization between the Serial Monitor readings and the ThingSpeak dashboard confirms that real-time data transmission to the cloud platform is functioning correctly. Overall, the results verify that the system can effectively detect hazardous gas conditions and respond immediately.

7.3 System Response Observation

During the gas exposure test, the system responded immediately once the sensor readings exceeded the predefined threshold value of 600. The red LED was activated and the buzzer was triggered automatically, indicating detection of hazardous gas levels. At the same time, the system status in the Serial Monitor changed to “GAS DETECTED!” and the data was successfully transmitted to the ThingSpeak cloud platform.

When the gas source was removed, the sensor readings gradually decreased below the threshold level. The system automatically deactivated the buzzer and switched back to safe mode, where only the green LED remained active.

This observation confirms that the system is capable of real-time detection, automatic alert activation, and stable recovery once environmental conditions return to normal.

8.3 System Performance Evaluation

Table 8.3 System Performance Evaluation

Condition	Average Reading	Threshold	Buzzer Status
Normal	~550-600	600	OFF
Gas Exposure	~685-761	600	ON

The difference between normal readings (approximately 550–600) and gas exposure readings (approximately 685–761) shows a significant increase of more than 100 units above the threshold value. This clear variation allows effective differentiation between safe and unsafe environmental conditions.

The system response time was observed to be within a few seconds after gas exposure. Once the gas source was removed, the readings gradually decreased below the threshold level of 600, and the buzzer automatically deactivated, returning the system to safe mode.

9. Industry Relevance and Future Trends

The Smart Leakage Monitoring System developed in this project demonstrates the practical use of IoT technology in safety monitoring applications. Gas detection systems are widely implemented in residential buildings, commercial kitchens, laboratories, and industrial facilities to prevent fire hazards and health risks caused by toxic gas exposure.

By integrating real-time monitoring, cloud data storage, and remote dashboard access, this system aligns with current industry practices in smart building and industrial safety management. The ability to monitor gas concentration remotely improves response efficiency and enhances preventive safety measures.

In the future, such systems can be enhanced with mobile notifications, AI-based predictive analysis, and integration into centralized building management systems. This reflects how IoT and cloud computing continue to play an important role in advancing intelligent and scalable safety monitoring solutions within the ICT industry.

10. Conclusion

In conclusion, the Smart Gas Leakage Monitoring System successfully achieved its objective of detecting gas levels in real time and providing immediate alerts when unsafe conditions were detected. The system was able to clearly differentiate between normal conditions (below the threshold of 600) and gas exposure conditions (above 600), ensuring reliable detection performance.

During testing, the system responded within a few seconds once the gas level exceeded the predefined threshold. The red LED and buzzer were activated automatically, while data was successfully transmitted to the ThingSpeak platform for monitoring purposes. When the gas level returned to normal, the system automatically switched back to safe mode, demonstrating stable recovery performance.

Overall, the project proves that IoT-based gas monitoring systems can enhance safety by providing early detection, real-time alerts, and remote monitoring capabilities.

11. Project Timeline (7 Weeks)

	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7
Planning & Feasibility Analysis							
Define project goals, scope, and objectives.							
Requirements Gathering & Analysis							
Document functional requirements and non-functional requirements							
System & Architecture Design							
Create circuit diagrams and select the communication protocol							
Enabling the wireless connection							
Development - Hardware & Embedded Coding							
Assemble physical hardware and write firmware to collect and process sensor data locally							
Perform unit tests on individual components like sensors or relays.							
Development - Cloud & Integration							
Connect devices to the cloud							
Develop the user dashboard for data visualization.							
Testing & Quality Assurance							
Perform integration testing to ensure hardware and cloud communicate reliably.							
Deployment & Maintenance							
Install the device in its final environment.							
Document the final setup and prepare a maintenance plan for future firmware updates.							